

Scientific Review & Synthesis

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A systematic review of the leaf traits considered to contribute to removal of airborne particulate matter pollution in urban areas

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1. Abstract & Scope

Vegetation removes particulate matter (PM) from the atmosphere, providing a crucial ecosystem service in urban areas. This systematic review synthesized global research on the leaf traits that contribute to PM removal. Coniferous needle leaves, waxy coatings, high trichome density, and narrow lanceolate leaf shapes were identified as key traits that promote PM capture. However, the review highlighted a profound lack of quantitative consensus across the literature due to methodological fragmentation (different washing techniques, gravimetric vs SEM vs image processing) and a failure to normalize background pollution levels.

2. Methodology & Experimental Parameters

Systematic search across Web of Science, Scopus, and PubMed databases. Categorization of leaf traits (coniferous needle, epicuticular wax, trichomes, leaf shape, petiole length) and critical analysis of research methods.

3. Core Findings & Data Synthesis

Needle leaves and waxy foliage consistently outperform flat broadleaves. Identified major research gaps, including lack of standardized extraction protocols and ignoring localized microclimate context (wind speed and humidity).

4. Strategic Relevance to JU and Kolkata Planning

Emphasizes the need to establish uniform surface normalization protocols ($\mu\text{g}/\text{cm}^2$) and warns against selecting urban planting species based on a single 'magic trait'.

